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NEW CORRELATIONS FOR CRUDE OIL PHYSICAL PROPERTIES

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SPE Members

Summary

The study presents nine new empirical correlations for crude oil properties: oil formation volume factor, solution gas-oil ratio, bubble point pressure, viscosity of gas-free oil, viscosity of gas saturated oil, viscosity of under saturated oil, isothermal compressibility of saturated oil, gas gravity correction factor, and conversion factor to convert the flash oil formation volume factor to the differential oil formation volume factor.

Recommendations for the use of the best correlation for a selected range of independent variables are also presented. In almost all ranges the new correlations provide the best predictions.

Introduction

Accuracy in estimating crude oil physical properties is of great importance to both the reservoir engineer and the production engineer. The reservoir engineer uses the fluid physical properties to calculate oil reserves and flow characteristics through porous media. The required correlation should be predominantly based on differential process, since it closely describes the gas and liquid flow in the reservoir. The production engineer, on the other hand, requires fluid physical properties based on flash liberation to calculate pressure drops in tubing and flowlines.

There are a number of empirical correlations used in the oil industry to predict reservoir fluid physical properties. However, the conditions under which these correlations should be used have never been thoroughly tested. In addition, some of these correlations were developed using flash data and some using differential data. Moses¹ has shown that the error using flash instead of differential data, or vice versa, can be as large as twenty percent.

The Data Bank

Wide Spread Source of Data

The data bank for this study was collected from PVT reports and literature. The first major source was from South East Asia, mainly Indonesia. The second source was North America

including the offshore area. The rest came from the Middle East and Latin America.

A set of 5392 data points was used to develop the correlations for flash oil formation volume factor- B_{of} , flash solution gas-oil ratio- R_{sf} , and bubble point pressure- p_b . These data represent 740 different crude oil samples. For the development of undersaturated oil properties correlations, i.e. c_o , μ_o , and B_{of} a total of 3588 data points collected from 661 different crude oil samples were used. A total of 661 samples was used to determine the dead oil viscosity- μ_{od} correlation and 5321 data points were used to establish live oil viscosity- μ_{ol} correlations. 208 data points were used to get the normalized separator gas gravity to a 100 psig separating condition. A set of 1802 data points was used to develop the conversion factor from differential liberation data to flash data.

The range of the variables, i.e. the flash gas-oil ratio- R_{sf} , API oil gravity, bubble point oil formation volume factor- B_{ofb} , separator gas gravity- γ_{gsep} , separator pressure- p_{sep} , separator temperature- T_{sep} , dead oil viscosity- μ_{od} , live oil viscosity- μ_{ol} , viscosity of undersaturated oil- μ_o , reservoir temperature- T , and reservoir pressure- p is listed in Table 1. According to Moses, these oils can be considered as ordinary oils and not condensate oils.

Collected data from the literature were not used in development of the new correlations, since insufficient information was available. However, these data were used to compare the existing correlations and the new correlations.

Flash Vaporization Data

Flash vaporization data used in this study were derived from two sources. The first set was obtained from PVT reports on the single stage flash separator tests of a reservoir fluid sample. The second source was obtained by converting the differential data from PVT reports to flash data using the following equations¹:

for the oil formation volume factor

$$B_{of} = B_{od} \frac{B_{ofb}}{B_{odb}}, \quad (1)$$

and for the solution gas-oil ratio

$$R_{sf} = R_{sfb} - (R_{sdb} - R_{sd}) \frac{B_{ofb}}{B_{odb}}. \quad (2)$$

Statistical Analysis

The correlations were developed using nonlinear regression analysis. The criteria used to select the best correlation were primarily based on the standard error of estimate. However, other statistical parameters such as *R-square*, average percent relative error and absolute percent relative error were also considered.

A sensitivity analysis was performed on all independent variables to determine under which condition a given correlation should be used. It was found that the new developed correlations, in almost all cases, out performed other fluid physical correlations.

PVT Correlations

Gas Specific Gravity Correction

The effect of separator pressure on fluid physical properties has been studied by Cook et al² and Vasquez³. There was a need to standardize the separator pressure at which the gas gravity is going to be reported. From the PVT reports of 208 points, following the approach of Vasquez it was found that the mean separator pressure is 101.6 psig. The reference pressure of 100 psig was therefore adopted. This is the same pressure that Vasquez-Beggs⁴ used in their report. Following the functional form of the Vasquez gas correction factor, the following empirical equation was obtained:

$$C = \frac{\gamma_{g100}}{\gamma_{gsep}} = 1.00 + 0.1595 \times API^{0.4078} \times T_{sep}^{-0.2466} \times \log \frac{P_{sep}}{114.7}. \quad (3)$$

Oil Formation Volume Factor

At and below bubble point pressure

Six empirical correlations, i.e. Standing⁵, Glaso⁶, Vasquez-Beggs, Obomanu-Okpobori⁷, Marhoun⁸, and Majeed-Salman⁹ were reviewed and evaluated using flash data bank. Since Standing's correlation was found to be the best when evaluated on the data bank, its functional form was used for the development of new correlation. The new correlation is given as:

$$B_{of} = 0.98496 + 0.0001 \times F^{1.50}, \quad (4)$$

where

$$F = R_{sf}^{0.755} \times \gamma_{g100}^{0.25} \times \gamma_o^{-1.50} + 0.45T. \quad (5)$$

Above bubble point pressure

For a given crude oil composition the relationship is given by:

$$B_{of} = B_{ofb} \times e^{c_o(p_b - p)}, \quad (6)$$

where B_{ofb} is the flash formation volume factor at the bubble point pressure that can be estimated by equation (4). The isothermal oil compressibility- c_o was calculated using equation (7).

Isothermal Oil Compressibility

Vasquez-Beggs and Calhoun¹⁰ have determined the isothermal expansion coefficient for crude oil, c_o . Several models have been tried, including the linear model proposed by Vasquez-Beggs to determine the best correlation for c_o . The final equation that was obtained is given by:

$$c_o = 6.8257 \times 10^{-6} \times R_{sf}^{0.5002} \times p \times T^{0.76606} \times \gamma_{g100}^{-0.35505}. \quad (7)$$

Solution Gas-Oil Ratio

Six correlations, i.e. Standing, Lasater¹¹, Glaso, Vasquez-Beggs, Marhoun, and Obomanu-Okpobori were reviewed and tested using the data bank. Even though Standing, Lasater, Glaso, and Marhoun developed their correlations for bubble point pressure prediction, their correlations can be rearranged to give equations for solution gas in oil at pressures below and at the original bubble point. This is possible because all conditions below the original bubble point also represent saturation conditions.

Since Vasquez-Beggs correlation was found to be the best of the existing correlations, it was decided to follow their approach and group the data into two groups. The new equation for the solution gas-oil ratio is given as:

for $API \leq 30$

$$R_{sf} = 0.05958 \times \gamma_{g100}^{0.7972} \times p^{1.0014} \times 10^{13.1405 API / (T+460)}, \quad (8)$$

for $API > 30$

$$R_{sf} = 0.03150 \times \gamma_{g100}^{0.7587} \times p^{1.0937} \times 10^{11.289 API / (T+460)}. \quad (9)$$

Bubble Point Pressure

Equations (8) and (9), that were developed for the solution gas-oil ratio, can be used to predict the bubble point pressure. Or, in other words, if a gas-oil ratio is known, which can either be the produced gas-oil ratio or the formation gas-oil ratio, the pressure at which this gas goes in solution can be calculated by inverting equations (8) and (9). The final form of bubble point pressure correlation is given as:

for $API \leq 30$

$$p = \left(\frac{R_{sf}}{0.05958 \times \gamma_{g100}^{0.7972} \times 10^{13.1405 API / (T+460)}} \right)^{0.9986}, \quad (10)$$

for $API > 30$

$$p = \left(\frac{R_{sf}}{0.03150 \times \gamma_{g100}^{0.7587} \times 10^{11.289API/(T+460)}} \right)^{0.9143} \quad (11)$$

Viscosity of Gas-Free Oil

The term dead oil viscosity represents the viscosity of gas-free oil. Among the popular dead oil viscosity correlations are those presented by Beal¹², Glaso, and Beggs-Robinson¹³. The basic independent variables are API and temperature. Glaso's functional form was adopted for the new dead oil viscosity correlation, since it gave the best result when tested using the data bank. The resultant equation is given as:

$$\mu_{od} = 16.0 \times 10^8 \times T^{-2.8177} \times (\log(API))^{5.7526 \times \log(T) - 26.9718} \quad (12)$$

Viscosity of Gas Saturated Oil

Live oil viscosity represents the viscosity of gas saturated oil system. The live oil viscosity correlations were developed by taking advantage of the fact that a linear relationship exists between $\log \mu_{od}$ and $\log \mu_{ol}$ for a particular value of dissolved gas. Two publications were considered in this study: the Chew-Connally's¹⁴ and the Beggs-Robinson's. The equation presented in this study is the Chew-Connally correlation, modified as:

$$\mu_{ol} = -0.06821 + 0.9824 \times f + 0.0004034 \times f^2, \quad (13)$$

where

$$f = (0.2001 + 0.8428 \times 10^{-0.000845 \times R_{sf}}) \times \mu_{od}^{(0.43 + 0.5165 \times y)}, \quad (14)$$

and

$$y = 10^{-0.00081 \times R_{sf}} \quad (15)$$

Viscosity of Undersaturated Oil

The only two correlations considered for predicting the undersaturated oil viscosity were those by Beal and Vasquez-Beggs. Sutton et al¹⁵ presented an empirical equation by curve fitting Beal's charts. The functional form of Sutton's was used in this study to develop a new correlation. The resultant equation is given as:

$$\mu_o = 1.00081 \times \mu_{ob} + 0.001127 \times (p - p_b) \times (-0.006517 \times \mu_{ob}^{1.8148} + 0.038 \times \mu_{ob}^{1.590}) \quad (16)$$

Conversion Factor from Flash Liberation Data to Differential Liberation

All the empirical equations developed in the proceeding sections are applicable only to a flash process. However, if a process is differential and a PVT report is not available, there might be a need to convert flash data to differential data. A flash to differential conversion factor is based only on field measurable

parameters such as tank oil gravity, separator gas gravity, separator pressure and temperature, and reservoir temperature. The final form of the conversion factor is given as:

$$\begin{aligned} C.F. &= \frac{B_{ofb}}{B_{odb}} \\ &= 0.7264 \times \gamma_o^{0.3202} - 0.3126 \times \gamma_{gsep}^{-0.02087} \\ &\quad + 0.6459 \times \left(\frac{T_{sep} + 460}{T + 460} \right)^{0.5596} \end{aligned} \quad (17)$$

This conversion factor- $C.F.$ can now be used to convert the B_{of} or B_{ofb} to the B_{od} or B_{odb} . Using equation (17) and equation (1) gives:

$$B_{od} = \frac{B_{of}}{C.F.}, \quad (18)$$

and

$$B_{odb} = \frac{B_{ofb}}{C.F.}, \quad (19)$$

where B_{of} or B_{ofb} can be calculated using equation (4).

However, the differential solution gas-oil ratio, R_{od} can not be calculated that way, since R_{odb} is not readily available.

Recommendations for the Use of Oil Physical Properties Correlations

The second purpose of this study is to recommend under which conditions a given correlation should be used. Until recently, the only guide for selecting a correlation was offered by Chierici et al¹⁵, who suggested the use of Lasater's correlation for oils with $API \geq 15$ and use of Standing's for oils with $API < 15$.

In order to be able to recommend the most appropriate correlation for a given set of independent variables (γ_o , γ_{gsep} , T_{sep} , p_{sep} , T , and p), the sensitivity analyses was performed. The results are summarized in Table 2.

Conclusions

The new empirical PVT correlations have been developed for gas specific gravity to a reference separator pressure, oil FVF, solution GOR, bubble point pressure, dead oil viscosity, live oil viscosity, undersaturated oil viscosity, isothermal compressibility, and conversion factor from flash liberation data to differential liberation data. Data used were from a number of locations worldwide and cover a broad range of oils.

Recommendations for the use of the best correlation for a selected range of independent variables are also presented. In almost all ranges the new correlations provide the best predictions.

The new correlations are developed for calculation of fluid properties using measured field surface data such as: gravity of separator gas, separator temperature and pressure, and API gravity of stock tank oil.

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Nomenclature

$^{\circ}API$	- Gravity of stock tank oil at 60°F.
B_{od}	- Barrels of oil at some reservoir pressure other than bubble point pressure required to yield one barrel residual oil at 60°F when differentially liberated to atmospheric pressure, <i>bbl/bbl residual oil</i> .
B_{odb}	- Barrels of bubble point oil required to yield one barrel residual oil at 60°F when differentially liberated to atmospheric pressure, <i>bbl/bbl resid. oil</i> .
B_{of}	- Barrels of oil at some reservoir pressure other than bubble point pressure required to yield one stock tank barrel oil at 60°F when flashed through separator to stock tank conditions, <i>bbl/STB</i> .
B_{ofb}	- Barrels of bubble point oil required to yield one stock tank barrel oil at 60°F when flashed through separator to stock tank conditions, <i>bbl/STB</i> .
C	- Correction factor for specific gravity of separator gas.
$C.F.$	- Conversion Factor for converting B_{of} or B_{ofb} to B_{od} or B_{odb} , <i>STB/bbl residual oil</i> .
c_o	- Isothermal compressibility, psi^{-1} .
F	- Correlating number for calculating oil FVF.
f	- Correlating number for calculating live oil viscosity.
$\log$	- Logarithm to the base 10.
p	- Reservoir pressure, <i>psia</i> .
p_b	- Bubble point pressure, <i>psia</i> .
p_{sep}	- Separator pressure, <i>psia</i> .
R_{od}	- Cubic feet of gas in solution at any pressure less than bubble point pressure in one barrel residual oil when measured by differential liberation, <i>SCF/bbl residual oil</i> .
R_{odb}	- Cubic feet of gas in solution at the bubble point pressure in one barrel residual oil when measured by differential liberation, <i>SCF/bbl residual oil</i> .
R_{of}	- Cubic feet of separator and stock tank gas at any pressure less than the bubble point pressure in one stock tank barrel oil, <i>SCF/STB</i> .
R_{ofb}	- Cubic feet of separator and stock tank gas at the bubble point pressure in one stock tank barrel oil, <i>SCF/STB</i> .
T	- Reservoir temperature, °F.
T_{sep}	- Separator temperature, °F.
y	- Correlating number for calculating live oil viscosity.
γ_{g100}	- Specific gravity of gas at T_{sep} and separation pressure of 100 psig (air=1).
γ_{gsep}	- Specific gravity of gas at T_{sep} and P_{sep} (air=1).
γ_o	- Specific gravity of stock tank oil at 60°F (water=1).
μ_o	- Viscosity of undersaturated oil, <i>cp</i> .
μ_{ob}	- Viscosity of oil at bubble point pressure, <i>cp</i> .
μ_{od}	- Viscosity of dead oil, <i>cp</i> .
μ_{ol}	- Viscosity of live oil, <i>cp</i> .

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Table 1

Measured Parameter Ranges

For B_{of} , R_{sf} , and p Correlations

Parameter	Minimum	Maximum
$^{\circ}API$	14.4	59.0
B_{od}	1.007	2.747
B_{ofb}	1.0003	2.144
B_{of}	1.007	2.144
p	14.7	6054.7
p_{sep}	14.7	1414.7
R_{sf}	0.0	2890.0
T	75.0	320.0
T_{sep}	38.0	294.0
γ_{gsep}	0.4824	1.668

For c_o and μ_o Correlations

Parameter	Minimum	Maximum
$^{\circ}API$	14.4	59.0
B_{od}	1.007	2.747
B_{ofb}	1.022	2.144
p	24.7	6014.7
p_b	24.7	4764.7
p_{sep}	14.7	1414.7
R_{sf}	0.0	2890.0
T	75.0	320.0
T_{sep}	38.0	294.0
γ_{gsep}	0.4824	1.668
μ_{ob}	0.168	184.86
μ_o	0.168	517.03

For μ_{od} and μ_{ol} Correlations

Parameter	Minimum	Maximum
$^{\circ}API$	14.4	59.0
B_{od}	1.007	2.747
B_{ofb}	1.022	2.144
p	24.7	7170.7
p_{sep}	14.7	1414.7
R_{sf}	0.0	2890.0
T	80.0	320.0
T_{sep}	38.0	294.0
γ_{gsep}	0.4824	1.668
μ_{od}	0.506	682.0
μ_{ol}	0.097	586.0

For C.F. Correlations

Parameter	Minimum	Maximum
$^{\circ}API$	15.2	57.6
B_{odb}	1.027	2.485
B_{ofb}	1.004	2.410
p_{sep}	14.7	1414.7
R_{sfb}	1.4	2473.0
T_{sep}	0.0	294.0
γ_{gsep}	0.4824	1.668

For C Correlations

Parameter	Minimum	Maximum
$^{\circ}API$	15.2	57.6
p_{sep}	14.7	514.7
R_{sf}	47.0	1505.0
T_{sep}	65.0	186.0
γ_{g100}	0.379	1.417
γ_{gsep}	0.379	1.709

Table 2.1

Sensitivity Analyses

Prediction of p , R_{sf} , and B_{of}

Using Flash Liberation Data (5392 samples)

p	R_{sf}	B_{of}
Model (263)*	Model (83)	Model (.0428)
Vasquez (264)	Vasquez (86)	Standing (.0451)
Standing (265)	Standing (86)	Marhoun (.0457)
Lasater (275)	Glaso (94)	Obomanu (.0463)
Glaso (298)	Obomanu (154)	Glaso (.0472)
Marhoun (387)	Marhoun (157)	Majeed (.0475)
Obomanu (401)	Lasater (245)	Vasquez (.0458)

Using Data from Literature (998 samples)

p	R_{sf}	B_{of}
Model (396)*	Model (171)	Model (.0527)
Vasquez (398)	Vasquez (175)	Vasquez (.0539)
Standing (407)	Standing (177)	Marhoun (.0546)
Lasater (409)	Glaso (179)	Majeed (.0581)
Glaso (413)	Lasater (232)	Standing (.0600)
Marhoun (511)	Obomanu (266)	Glaso (.0614)
Obomanu (544)	Marhoun (287)	Obomanu (.0650)

*: numbers in brackets indicate the standard error of estimate

Number of samples from literature:

Glaso = 45

Rzasa-Borden¹⁸ = 8Katz¹⁶ = 53

Majeed-Salman = 420

Obomanu-Okpobori = 48

Vasquez = 256

Osterman¹⁷ = 8

Marhoun = 160

Effect of API Oil GravityAPI < 20 (128 samples)

p	R_{sf}	B_{of}
1. Model	Model	Glaso
2. Vasquez	Vasquez	Standing
3. Standing	Standing	Model
4. Lasater	Glaso	Marhoun
5. Glaso	Marhoun	Obomanu
6. Obomanu	Obomanu	Vasquez
7. Marhoun	Lasater	Majeed

Effect of γ_{gsep} $\gamma_{gsep} < 0.60$ (529 samples)

p	R_{sf}	B_{of}
1. Lasater	Model	Standing
2. Standing	Vasquez	Glaso
3. Vasquez	Standing	Model
4. Model	Glaso	Vasquez
5. Glaso	Obomanu	Majeed
6. Marhoun	Marhoun	Marhoun
7. Obomanu	Lasater	Obomanu

20 < API < 30 (796 samples)

p	R_{sf}	B_{of}
1. Lasater	Model	Model
2. Standing	Vasquez	Marhoun
3. Vasquez	Standing	Standing
4. Model	Glaso	Vasquez
5. Glaso	Obomanu	Glaso
6. Obomanu	Marhoun	Obomanu
7. Marhoun	Lasater	Majeed

0.60 < γ_{gsep} < 0.80 (2393 samples)

p	R_{sf}	B_{of}
1. Model	Model	Model
2. Standing	Vasquez	Majeed
3. Vasquez	Standing	Marhoun
4. Lasater	Marhoun	Obomanu
5. Marhoun	Glaso	Vasquez
6. Glaso	Obomanu	Standing
7. Obomanu	Lasater	Glaso

30 < API < 45 (3863 samples)

p	R_{sf}	B_{of}
1. Model	Model	Model
2. Lasater	Vasquez	Standing
3. Vasquez	Standing	Marhoun
4. Standing	Glaso	Glaso
5. Glaso	Marhoun	Obomanu
6. Marhoun	Obomanu	Majeed
7. Obomanu	Lasater	Vasquez

0.80 < γ_{gsep} < 1.00 (2005 samples)

p	R_{sf}	B_{of}
1. Model	Model	Model
2. Standing	Vasquez	Standing
3. Vasquez	Standing	Marhoun
4. Lasater	Glaso	Vasquez
5. Glaso	Obomanu	Obomanu
6. Marhoun	Marhoun	Majeed
7. Obomanu	Lasater	Glaso

API > 45 (605 samples)

p	R_{sf}	B_{of}
1. Model	Model	Model
2. Standing	Standing	Marhoun
3. Vasquez	Vasquez	Standing
4. Lasater	Glaso	Majeed
5. Marhoun	Marhoun	Vasquez
6. Glaso	Obomanu	Glaso
7. Obomanu	Lasater	Obomanu

1.00 < γ_{gsep} < 1.50 (529 samples)

p	R_{sf}	B_{of}
1. Lasater	Glaso	Model
2. Glaso	Standing	Marhoun
3. Vasquez	Vasquez	Vasquez
4. Standing	Marhoun	Majeed
5. Model	Model	Obomanu
6. Marhoun	Obomanu	Standing
7. Obomanu	Lasater	Glaso

$\gamma_{resp} > 1.50$ (10 samples)

p	R_{sf}	B_{of}
1. Obomanu	Obomanu	Marhoun
2. Glaso	Glaso	Model
3. Lasater	Model	Vasquez
4. Vasquez	Standing	Majeed
5. Standing	Vasquez	Obomanu
6. Model	Marhoun	Standing
7. Marhoun	Lasater	Glaso

 $T > 300$ (17 samples)

p	R_{sf}	B_{of}
1. Marhoun	Marhoun	Model
2. Obomanu	Obomanu	Marhoun
3. Glaso	Lasater	Glaso
4. Vasquez	Glaso	Majeed
5. Model	Vasquez	Standing
6. Standing	Standing	Vasquez
7. Lasater	Model	Obomanu

Effect of Temperature $T < 100$ (242 samples)

p	R_{sf}	B_{of}
1. Model	Model	Glaso
2. Marhoun	Glaso	Model
3. Vasquez	Vasquez	Standing
4. Standing	Standing	Marhoun
5. Lasater	Obomanu	Majeed
6. Glaso	Marhoun	Vasquez
7. Obomanu	Lasater	Obomanu

 $100 < T < 200$ (3066 samples)

p	R_{sf}	B_{of}
1. Lasater	Model	Model
2. Model	Vasquez	Standing
3. Vasquez	Standing	Marhoun
4. Standing	Glaso	Obomanu
5. Glaso	Marhoun	Glaso
6. Marhoun	Obomanu	Majeed
7. Obomanu	Lasater	Vasquez

 $200 < T < 300$ (2163 samples)

p	R_{sf}	B_{of}
1. Lasater	Model	Model
2. Model	Vasquez	Marhoun
3. Vasquez	Standing	Standing
4. Standing	Glaso	Glaso
5. Glaso	Marhoun	Obomanu
6. Marhoun	Obomanu	Majeed
7. Obomanu	Lasater	Vasquez

Table 2.2**Sensitivity Analyses****Prediction of μ_o (3588 samples)**

1. Model (*standard error of estimate* = 6.71)
2. Beal (*standard error of estimate* = 7.55)
3. Vasquez (*standard error of estimate* = 8.96)

Effect of p_b on prediction of μ_o

$p_b \leq 515$ 1012 samples	$515 < p_b \leq 1015$ 433 samples	$1015 < p_b \leq 1515$ 638 samples
1. Model	Model	Vasquez
2. Beal	Beal	Model
3. Vasquez	Vasquez	Beal

$1515 < p_b \leq 2015$ 475 samples	$2015 < p_b \leq 3015$ 661 samples
1. Model	Vasquez
2. Beal	Model
3. Vasquez	Beal

$3015 < p_b \leq 4015$ 303 samples	$p_b > 4015$ 166 samples
1. Model	Vasquez
2. Beal	Beal
3. Vasquez	Model

Table 2.3
Sensitivity Analyses

Prediction of μ_{od} (661 samples)

1. Model (<i>standard error of estimate</i> = 18.84)
2. Glaso (<i>standard error of estimate</i> = 19.20)
3. Beal (<i>standard error of estimate</i> = 20.72)
3. Beggs-Robinson (<i>standard error of estimate</i> = 36.98)

Effect of T on prediction of μ_{od}

$T \leq 100$ 13 samples	$100 < T \leq 200$ 381 samples
1. Beggs 2. Model 3. Glaso 4. Beal	Model Glaso Beal Beggs

$200 < T \leq 300$ 260 samples	$T > 300$ 7 samples
1. Model 2. Glaso 3. Beal 4. Beggs	Glaso Beggs Beal Model

Table 2.4
Sensitivity Analyses

Prediction of μ_{ol} (5321 samples)

1. Model (<i>standard error of estimate</i> = 3.96)
2. Chew-Connally (<i>standard error of estimate</i> = 4.05)
3. Beggs-Robinson (<i>standard error of estimate</i> = 4.14)

Effect of R_{sf} on prediction of μ_{ol}

$R_{sf} \leq 50$ 343 samples	$50 < R_{sf} \leq 100$ 459 samples	$100 < R_{sf} \leq 250$ 1406 samples
1. Model 2. Beggs 3. Chew	Chew Model Beggs	Model Beggs Chew

$250 < R_{sf} \leq 500$ 1598 samples	$500 < R_{sf} \leq 1000$ 1168 samples	$1000 < R_{sf} \leq 1500$ 250 samples
1. Model 2. Beggs 3. Chew	Model Chew Beggs	Model Chew Beggs

$1500 < R_{sf} \leq 2000$ 72 samples	$R_{sf} > 2000$ 23 samples
1. Model 2. Beggs 3. Chew	Beggs Chew Model

Table 2.5
Sensitivity Analyses

Prediction of c_o (2545 samples)

1. Model (<i>average percent relative error</i> * = 23.67)
2. Calhoun (<i>average percent relative error</i> = 70.21)
3. Vasquez (<i>average percent relative error</i> = 87.32)

* : since *standard error of estimate* is very small

Effect of API on prediction of c_o

$API \leq 20$ 168 samples	$20 < API \leq 30$ 411 samples
1. Model 2. Vasquez 3. Calhoun	Model Vasquez Calhoun

$30 < API \leq 45$ 1827 samples	$API > 45$ 139 samples
1. Model 2. Vasquez 3. Calhoun	Vasquez Model Calhoun

Table 2.6**Sensitivity Analyses****Prediction of B_{of} (2545 samples)****(Undersaturated oil FVF)**

1. Model (<i>standard error of estimate</i> = 0.00551)
2. Vasquez (<i>standard error of estimate</i> = 0.00606)
3. Calhoun (<i>standard error of estimate</i> = 0.00777)

Effect of API on prediction of B_{of}

$API \leq 20$ 168 samples	$20 < API \leq 30$ 411 samples
1. Calhoun	Model
2. Model	Vasquez
3. Vasquez	Calhoun

$30 < API \leq 45$ 1827 samples	$API > 45$ 139 samples
1. Model	Vasquez
2. Vasquez	Model
3. Calhoun	Calhoun

Table 2.7**Sensitivity Analyses****Prediction of Gas Gravity Correction— C (208 samples)**

1. Model (<i>standard error of estimate</i> = 0.04767)
2. Vasquez (<i>standard error of estimate</i> = 0.05889)

Table 2.8**Sensitivity Analyses****Prediction of Conversion Factor****From Flash Data to Differential Data— $C.F.$ (1802 samples)**

Model (<i>standard error of estimate</i> = 0.03108)
--

```

C *****
C
C THIS PROGRAM CALCULATES THE SOLUTION GAS-OIL RATIO, THE BUBBLE
C POINT PRESSURE, THE OIL FORMATION VOLUME FACTOR, THE DEAD OIL VIS-
C COSITY, THE LIVE OIL VISCOSITY AND THE ADJUSTMENT FACTOR FOR CON-
C VERTING DATA FROM FLASH TO DIFFERENTIAL LIBERATION CONDITIONS.
C
C Written by:  R.S. Trijana Kartoatmodjo                      June 1991
C
C      * *  TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS  * *
C
C *****
C
C This program calculates specific gravity of gas corrected to 100
C psig separation condition using SUBROUTINE RSTGASC. It calculates
C the solution gas-oil ratio using SUBROUTINE RSTRSOL and calculates
C the bubble point pressure using SUBROUTINE RSTBPP.
C
C The isothermal oil compressibility is calculated using SUBROUTINE
C RSTCOMP. The oil formation volume factor is calculated for several
C conditions i.e. below bubble point or at bubble point or above
C bubble point pressure using SUBROUTINE RSTOFVF.
C
C The program calculates dead oil viscosity using SUBROUTINE RSTVISD.
C To calculate the live oil viscosity we have to considered the stage
C of pressure i.e. below bubble point, at bubble point or above
C bubble point pressure before using SUBROUTINE RSTVISO.
C
C This program also calculates the adjustment factor for converting
C oil formation volume factor from flash liberation to differential
C liberation conditions using SUBROUTINE RSTCONV.
C
C The English system of units is used in the calculation.
C
C *****
C
C      INPUT/OUTPUT LOGICAL FILE VARIABLES
C      -----
C
C IOERR = Output file for error messages when input values
C         passed to the subroutine are out of range.
C
C
C      VARIABLE DESCRIPTION
C      -----
C
C *API   = API gravity of the liquid. (deg)
C BO      = Oil formation volume factor. (bbl/STBO)
C BOFB    = Oil formation volume factor at the bubble point
C          pressure by flash liberation. (bbl/STBO)
C BOD     = Barrels of oil at some pressure other than bubble
C          point pressure required to yield 1 residual oil at
C          60 deg-F when differentially liberated to atmospheric
C          pressure, (bbl/bbl)
C CO      = Oil compressibility. (psi**-1)
C IERR    = Error code. (0=OK, 1=input variables out of range)
C IOERR   = Output file for error messages when input values
C          passed to the subroutine are out of range.
C
C *P      = Pressure. (psia)
C PB      = Bubble point pressure. (psia)
C *PSEP   = Separator pressure at which specific gravity of
C          produced gas is reported. (psia)
C
C *R      = Total producing gas/oil ratio. (scf/STBO)

```

```

C          = Separator gas/oil ratio + Tank oil gas/oil ratio.
C          (scf/STBO)
C      RS    = Solution gas/oil ratio. (scf/STBO)
C      RSD    = Solution gas/oil ratio in 1 bbl residual oil at any
C              pressure less than bubble point pressure when measured
C              by differential liberation. (scf/bbl)
C      SGO    = Specific gravity of oil.
C      *SGPG  = Specific gravity of separator gas.
C      SG100  = Specific gravity of gas at 100 psig and T-separator.
C      *T     = Temperature. (deg-F)
C      *TSEP  = Separator temperature at which specific gravity of
C              produced gas is reported. (deg-F)
C      VISD   = Dead oil viscosity. (cp)
C      VISO   = Live oil viscosity. (cp)
C      (* Indicates input variables)
C
C      *****
C
C      Program RSTKCOR
C      Character*1 CTEMP(80)
C
C      Print *, ' '
C      Print *, '*****'
C      Print *, '&'
C      Print *, '& WELCOME TO THE NEW CORRELATIONS FOR ESTIMATING &'
C      Print *, '& HYDROCARBON LIQUID PROPERTIES USING THE FOLLOW- &'
C      Print *, '& ING FIELD PARAMETERS: &'
C      Print *, '& 1. API TANK OIL GRAVITY, &'
C      Print *, '& 2. SEPARATOR GAS GRAVITY, &'
C      Print *, '& 3. SEPARATOR TEMPERATURE (DEG-F), &'
C      Print *, '& 4. SEPARATOR PRESSURE (PSIA), &'
C      Print *, '& 5. TOTAL PRODUCTION GAS OIL RATIO (SCF/STB), &'
C      Print *, '& 6. INSITU PRESSURE (PSIA), AND &'
C      Print *, '& 7. INSITU TEMPERATURE (DEG-F). &'
C      Print *, '&'
C      Print *, '& R.S. Trijana Kartoatmodjo and Zelimir Schmidt: &'
C      Print *, '& "NEW CORRELATIONS FOR &'
C      Print *, '& CRUDE OIL PHYSICAL PROPERTIES" &'
C      Print *, '& SPE 023556 &'
C      Print *, '& June 20, 1991 &'
C      Print *, '&'
C      Print *, '&'
C      Print *, '*****'
C      Print *, ' '
C      Print *, ' '
C      Print *, ' '
C
C      Open (7,File='JUNK',Status='UNKNOWN')
C      Print *, 'PRESS RETURN TO CONTINUE'
C      Read *,
C
C      10 Print *, 'ENTER THE VALUE OF STOCK TANK OIL GRAVITY (API)----->'
C      Read (6,5000) CTEMP
C      Do 20 K = 1, 80
C      If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
C      * CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
C      * CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
C      * CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
C      * Then
C      Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
C      Print *, 'PLEASE TRY AGAIN'
C      Print *, ' '
C      Go To 10
C      End If

```

```

20 Continue
  Write (7,5000) (CTEMP(I),I = 1,80)
  Rewind (7)
  Read (7,*) API
  Rewind (7)
C
  If (API.GT.70.0.OR.API.LT.10.0) Then
    Print *, 'THE API OIL GRAVITY IS OUT OF RANGE '
    Print *, 'PLEASE TRY AGAIN'
    Go To 10
  End If
C
30 Print *, 'ENTER THE VALUE OF SPECIFIC GRAVITY OF SEPARATOR GAS'
  Print *, ' (AIR=1) ----->'
  Read (6,5100) CTEMP
  Do 40 K = 1, 80
    If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
*      CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
*      CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
*      CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
*      Then
        Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
        Print *, 'PLEASE TRY AGAIN'
        Print *, ' '
        Go To 30
      End If
40 Continue
  Write (7,5100) (CTEMP(I),I = 1,80)
  Rewind (7)
  Read (7,*) SGPG
  Rewind (7)
C
  If (SGPG.GT.4.0.OR.SGPG.LT.0.4) Then
    Print *, 'THE SPECIFIC GRAVITY OF SEPARATOR GAS IS OUT OF RANGE'
    Print *, 'PLEASE TRY AGAIN'
    Go To 30
  End If
C
50 Print *, 'ENTER THE VALUE OF SEPARATOR TEMPERATURE (DEG-F) ---->'
  Read (6,5200) CTEMP
  Do 60 K = 1, 80
    If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
*      CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
*      CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
*      CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
*      Then
        Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
        Print *, 'PLEASE TRY AGAIN'
        Print *, ' '
        Go To 50
      End If
60 Continue
  Write (7,5200) (CTEMP(I),I = 1,80)
  Rewind (7)
  Read (7,*) TSEP
  Rewind (7)
C
  If (TSEP.GT.400.0.OR.TSEP.LT.-50.0) Then
    Print *, 'THE SEPARATOR TEMPERATURE IS OUT OF RANGE'
    Print *, 'PLEASE TRY AGAIN'
    Go To 50
  End If
C
70 Print *, 'ENTER THE VALUE OF SEPARATOR PRESSURE (PSIA) ----->'

```

```

Read (6,5300) CTEMP
Do 80 K = 1, 80
  If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
*    CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
*    CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
*    CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
*    Then
      Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
      Print *, 'PLEASE TRY AGAIN'
      Print *, ' '
      Go To 70
    End If
80 Continue
Write (7,5300) (CTEMP(I),I = 1,80)
Rewind (7)
Read (7,*) PSEP
Rewind (7)
C
  If (PSEP.GT.7000.0.OR.PSEP.LT.0.0) Then
    Print *, 'THE SEPARATOR PRESSURE IS OUT OF RANGE'
    Print *, 'PLEASE TRY AGAIN'
    Go To 70
  End If
C
90 Print *, 'ENTER THE TOTAL PRODUCED GAS OIL RATIO (SCF/STB) ---->'
Read (6,5400) CTEMP
Do 100 K = 1, 80
  If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
*    CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
*    CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
*    CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
*    Then
      Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
      Print *, 'PLEASE TRY AGAIN'
      Print *, ' '
      Go To 90
    End If
100 Continue
Write (7,5400) (CTEMP(I),I = 1,80)
Rewind (7)
Read (7,*) R
Rewind (7)
C
  If (R.GT.8000.0.OR.R.LT.0.0) Then
    Print *, 'THE TOTAL PRODUCED GAS OIL RATIO IS OUT OF RANGE'
    Print *, 'PLEASE TRY AGAIN'
    Go To 90
  End If
C
110 Print *, 'ENTER THE VALUE OF INSITU PRESSURE (PSIA) ----->'
Read (6,5500) CTEMP
Do 120 K = 1, 80
  If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
*    CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
*    CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
*    CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
*    Then
      Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
      Print *, 'PLEASE TRY AGAIN'
      Print *, ' '
      Go To 110
    End If
120 Continue
Write (7,5500) (CTEMP(I),I = 1,80)

```

```

Rewind (7)
Read (7,*) P
Rewind (7)
C
  If (P.GT.15000.OR.P.LT.0.0) Then
    Print *, 'THE INSITU PRESSURE IS OUT OF RANGE'
    Print *, 'PLEASE TRY AGAIN'
    Go To 110
  End If
C
130 Print *, 'ENTER THE VALUE OF INSITU TEMPERATURE (DEG-F) ----->'
    Read (6,5600) CTEMP
    Do 140 K = 1, 80
      If (CTEMP(K).NE.'0'.AND.CTEMP(K).NE.'1'.AND.CTEMP(K).NE.'2'.AND.
*      CTEMP(K).NE.'3'.AND.CTEMP(K).NE.'4'.AND.CTEMP(K).NE.'5'.AND.
*      CTEMP(K).NE.'6'.AND.CTEMP(K).NE.'7'.AND.CTEMP(K).NE.'8'.AND.
*      CTEMP(K).NE.'9'.AND.CTEMP(K).NE.'.'.AND.CTEMP(K).NE.' ')
*      Then
        Print *, 'INVALID INPUT VALUE--THE INPUT MUST BE A NUMBER'
        Print *, 'PLEASE TRY AGAIN'
        Print *, ' '
        Go To 130
      End If
140 Continue
    Write (7,5600) (CTEMP(I),I = 1,80)
    Rewind (7)
    Read (7,*) T
    Rewind (7)
C
  If (T.GT.1000.OR.T.LT.0.0) Then
    Print *, 'THE INSITU TEMPERATURE IS OUT OF RANGE'
    Print *, 'PLEASE TRY AGAIN'
    Go To 130
  End If
C
Print *, 'PRESS RETURN TO CONTINUE'
Read *,
C
Open (Unit=108,File='pvt.res')
C
*****
C
Call the subroutine
C
*****
C
-----
C
Calculate gravity gas at 114.7 psia.
C
-----
C
Call RSTGASC(SGPG,PSEP,TSEP,API,SG100,IOERR,IERR)
C
-----
C
Calculate Solution gas oil ratio.
C
-----
C
Call RSTRSOL(P,T,API,R,SG100,RS,IOERR,IERR)
C
-----
C
Calculate bubble point pressure.
C
-----
C
Call RSTBPP(T,API,R,SG100,PB,IOERR,IERR)
C
-----
C
Calculate isothermal oil compressibility.
C
-----
C
Call RSTCOMP(T,P,API,SG100,R,CO,IOERR,IERR)
C
-----
C
Calculate oil formation volume factor.
C
-----
C
Call RSTOFVF(T,P,API,SG100,PB,R,CO,BO,RS,IOERR,IERR)

```



```

Print *, 'OIL COMPRESSIBILITY, 1/PSIA           = ', CO
Print *, ' '
Print *, 'DEAD OIL VISCOSITY, CP                = ', VISD
Print *, ' '
Print *, 'LIVE OIL VISCOSITY, CP                = ', VISO
Print *, ' '
Print *, 'DIFFERENTIAL LIBERATION OIL FVF, bbl/BBL = ', BOD
Print *, ' '
Print *, ' '
Print *, 'PRESS RETURN TO CONTINUE '
Read *,
Print *, ' '
Print *, 'ADJUSTMENT FACTOR, STB/BBL            = ', CONV
Print *, ' '
Print *, 'FLASH OIL FVF AT BPP, bbl/STB         = ', BOFB
Print *, ' '
Print *, 'DIFFERENTIAL OIL FVF AT BPP, bbl/BBL   = ', BODB
Print *, ' '
Print *, 'PRESS RETURN TO CONTINUE '
Read *,
Print *, ' '
Print *, ' '
Print *, 'WHERE:'
Print *, ' '
Print *, 'bbl = BARREL OF OIL AT INSITU CONDITIONS'
Print *, ' '
Print *, 'BBL = BARREL OF RESIDUAL OIL'
Print *, ' '
Print *, 'BPP = BUBBLE POINT PRESSURE'
Print *, ' '
Print *, 'FVF = FORMATION VOLUME FACTOR'
Print *, ' '
Print *, 'GOR = GAS OIL RATIO'
Print *, ' '
Print *, 'STB = BARREL OF STOCK TANK OIL'
Print *, ' '

```

C

```

Print *, 'PRESS RETURN TO CONTINUE'
Read *,

```

C

```

If (PB.GT.P) Then
  Print *, ' '
  Print *, '#####'
  Print *, '&'
  Print *, '& NOTE:'
  Print *, '&   THE INSITU CONDITION OF THE OIL SYSTEM IS'
  Print *, '&   A SATURATED OIL CONDITION.'
  Print *, '&'
  Print *, '#####'
Else If (PB.EQ.P) Then
  Print *, ' '
  Print *, '#####'
  Print *, '&'
  Print *, '& NOTE:'
  Print *, '&   THE INSITU CONDITION OF THE OIL SYSTEM IS'
  Print *, '&   AT BUBBLE POINT PRESSURE CONDITION.'
  Print *, '&'
  Print *, '#####'
Else
  Print *, ' '
  Print *, '#####'
  Print *, '&'
  Print *, '& NOTE:'
  Print *, '&   THE INSITU CONDITION OF THE OIL SYSTEM IS'

```

```

      Print *, '&      AN UNDERSATURATED OIL CONDITION.      &'
      Print *, '&      &'
      Print *, '&      &'
End If

C
C
C *****
C Print output.
C *****
C
      Write (108,5700)
      Write (108,5800) P, T, SGPG, R, API, TSEP, PSEP
      Write (108,5900)
      Write (108,6000) SG100, RS, PB, BO, CO, VISD, VISO, BOD, CONV,
*      BOFB, BODB
      Stop
5000 Format (80A)
5100 Format (80A)
5200 Format (80A)
5300 Format (80A)
5400 Format (80A)
5500 Format (80A)
5600 Format (80A)
5700 Format (1X,'INPUT DATA',///)
5800 Format (1X,'IN SITU PRESSURE, PSIA',T40,'=',F15.4,/,1X,
*      'IN SITU TEMPERATURE, DEG. F',T40,'=',F15.4,/,1X,
*      'SEPARATOR GAS SPECIFIC GRAVITY',T40,'=',F15.4,/,1X,
*      'TOTAL PRODUCED GOR SCF/STB',T40,'=',F15.4,/,1X,
*      'API TANK OIL GRAVITY',T40,'=',F15.4,/,1X,
*      'SEPARATOR TEMPERATURE, DEG. F',T40,'=',F15.4,/,1X,
*      'SEPARATOR PRESSURE, PSIA',T40,'=',F15.4,///)
5900 Format (1X,'OUTPUT DATA',///)
6000 Format (1X,'GAS SPECIFIC GRAVITY AT 100 PSIG',T40,'=',F15.4,/,1X,
*      'FLASH SOLUTION GOR, SCF/STB',T40,'=',F15.4,/,1X,
*      'BUBBLE POINT PRESSURE, PSIA',T40,'=',F15.4,/,1X,
*      'FLASH OIL FVF, bbl/STB',T40,'=',F15.4,/,1X,
*      'OIL COMPRESSIBILITY, 1/PSIA',T40,'=',F15.4,/,1X,
*      'DEAD OIL VISCOSITY, CP',T40,'=',F15.4,/,1X,
*      'LIVE OIL VISCOSITY, CP',T40,'=',F15.4,/,1X,
*      'DIFFERENTIAL OIL FVF, bbl/BBL',T40,'=',F15.4,/,1X,
*      'ADJUSTMENT FACTOR STB/BBL',T40,'=',F15.4,/,1X,
*      'FLASH OIL FVF AT BBP, bbl/STB',T40,'=',F15.4,/,1X,
*      'DIFFERENTIAL OIL FVF AT BPP, bbl/BBL',T40,'=',F15.4,/)

C
      End

C
C *****
C      THIS SUBROUTINE CALCULATES GAS GRAVITY CORRECTION FACTOR
C      Written by: R.S. Trijana Kartoatmodjo      June 1991
C      * * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *
C      *****
C
      This program calculates specific gravity of gas at separator pres-
      sure of 100 psig using the commonly measured parameter in the oil
      field such as separator pressure and temperature at which specific
      gravity of produced gas is reported, specific gravity of the sepa-
      rator gas, and the API gravity of stock tank oil.

      The English system of units is used in the calculation.

```

REFERENCES

- C 1. Kartoatmodjo, R.S.T., Schmidt, Z.: ``New Correlations for Crude
C Oil Physical Properties,`` SPE 023556, June 1991
C
C 2. Kartoatmodjo, R.S.T.,: ``New Correlations for Estimating
C Hydrocarbon Liquid Physical Properties,`` M.Thesis, University
C of Tulsa, 1990.
C
C

C *****
C

C INPUT/OUTPUT LOGICAL FILE VARIABLES
C -----
C

C IOERR = Output file for error messages when input values
C passed to the subroutine are out of range.
C
C

C VARIABLE DESCRIPTION
C -----
C

C *API = API gravity of the stock tank oil. (deg)
C CORSG = Correction factor to gas gravity at 114.7 psia.
C IERR = Error code. (0=OK, 1=input variables out of range)
C IOERR = Output file for error messages when input values
C passed to the subroutine are out of range.
C *PSEP = Separator pressure at which specific gravity of
C produced gas is reported. (psia)
C *SGPG = Specific gravity of produced gas.
C SG100 = Specific gravity of gas at 100 psig and T.
C *TSEP = Separator temperature at which specific gravity of
C produced gas is reported. (deg-F)
C (* Indicates input variables)
C
C

C *****
C

C Subroutine RSTGASC(SGPG,PSEP,TSEP,API,SG100,IOERR,IERR)
C
C

C *****
C

C Check input variables for valid range.
C *****
C

C IERR = 0
C If (SGPG.LT.0.50.OR.SGPG.GE.3.0) Then
C Write (IOERR,*) ': Illegal input value for SGPG'
C IERR = 1
C Else If (PSEP.LT.0.0.OR.PSEP.GT.6000.0) Then
C Write (IOERR,*) ': Illegal input value for PSEP'
C IERR = 1
C Else If (TSEP.LT.0.0.OR.TSEP.GT.400.0) Then
C Write (IOERR,*) ': Illegal input value for TSEP'
C IERR = 1
C Else If (API.LE.0.0.OR.API.GT.100.0) Then
C Write (IOERR,*) ': Illegal input value for API'
C IERR = 1
C End If
C If (IERR.NE.1) Then
C
C

C -----
C Check input variables for valid correlation range.
C -----
C

C If (SGPG.LT.0.50.OR.SGPG.GT.2.26) Then
C Write (IOERR,*) ': SGPG out of range for ',
C * ' correlation, extrapolation occurring'

```

      Else If (PSEP.LT.0.0.OR.PSEP.GT.1800.0) Then
        Write (IOERR,*) ': PSEP out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (TSEP.LT.38.0.OR.TSEP.GT.294.0) Then
        Write (IOERR,*) ': TSEP out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (API.LT.14.4.OR.API.GT.58.95) Then
        Write (IOERR,*) ': API out of range for ',
*      ' correlation, extrapolation occurring'
      End If

```

```

*****
End of validity check.
*****

```

```

-----
Calculate gas gravity at 114.7 psia.
-----

```

```

      CORSG = 1. + 0.1595 * (API**0.4078) * (TSEP**(-0.2466)) *
*      ALOG10(PSEP/114.7)
      SG100 = CORSG * SGPG
End If

```

```

*****
END OF GAS GRAVITY CORRECTION TO 114.7 PSIA SEPARATOR PRESSURE.
*****

```

```

Return
End

```

```

*****
      THIS SUBROUTINE CALCULATES SOLUTION GAS-OIL RATIO
Written by: R.S. Trijana Kartoatmodjo June 1991
* * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *
*****

```

```

This program calculates solution gas/oil ratio using the commonly
measured parameters in the oil fields such as API gravity of stock
tank oil, total producing gas oil/ratio, reservoir temperature, re-
servoir pressure, and the output from SUBROUTINE RSTGASC.

```

```

The English system of units is used in the calculation.

```

REFERENCES

1. Kartoatmodjo, R.S.T., Schmidt, Z.: ``New Correlations for Crude Oil Physical Properties,`` SPE 023556, June 1991
2. Kartoatmodjo, R.S.T.: ``New Correlations for Estimating Hydrocarbon Liquid Physical Properties,`` M.Thesis, University of Tulsa, 1990.

```

*****

```

INPUT/OUTPUT LOGICAL FILE VARIABLES

```

IOERR = Output file for error messages when input values
passed to the subroutine are out of range.

```

VARIABLE DESCRIPTION

```

C
C      *API      = API gravity of the stock tank oil. (deg)
C      IERR      = Error code. (0=OK, 1=input variables out of range)
C      IOERR     = Output file for error messages when input values
C                  passed to the subroutine are out of range.
C      *P        = Pressure. (psia)
C      *R        = Total producing gas/oil ratio. (scf/STBO)
C                  = Separator gas/oil ratio + Tank oil gas/oil ratio.
C                  (scf/STBO)
C      RS        = Solution gas/oil ratio. (scf/STBO)
C      *SG100    = Specific gravity of gas at 100 psig and T-separator.
C                  As an input data or calculated from SUBROUTINE RSTGASC.
C      *T        = Temperature. (deg-F)
C      TABS      = Absolute temperature. (deg-R)
C      A,B,C,D= Dummy variables.
C      (* Indicates input variables)
C
C      *****
C
C      Subroutine RSTRSOL(P,T,API,R,SG100,RS,IOERR,IERR)
C
C      *****
C      Check input variables for valid range.
C      *****
C
C      IERR = 0
C      If (P.LT.0.0.OR.P.GT.9000.0) Then
C        Write (IOERR,*) ': Illegal input value for P'
C        IERR = 1
C      Else If (T.LT.0.0.OR.T.GT.400.0) Then
C        Write (IOERR,*) ': Illegal input value for T'
C        IERR = 1
C      Else If (SG100.LT.0.50.OR.SG100.GE.3.0) Then
C        Write (IOERR,*) ': Illegal input value for SGPG'
C        IERR = 1
C      Else If (API.LE.0.0.OR.API.GT.100.0) Then
C        Write (IOERR,*) ': Illegal input value for API'
C        IERR = 1
C      Else If (R.LT.0.0) Then
C        Write (IOERR,*) ': Illegal input value for RTO'
C        IERR = 1
C      End If
C      If (IERR.NE.1) Then
C
C      -----
C      Check input variables for valid correlation range.
C      -----
C
C      If (P.LT.0.0.OR.P.GT.6040.0) Then
C        Write (IOERR,*) ': P out of range for ',
C      *      ' correlation, extrapolation occurring'
C      Else If (T.LT.75.0.OR.T.GT.320.0) Then
C        Write (IOERR,*) ': T out of range for ',
C      *      ' correlation, extrapolation occurring'
C      Else If (SG100.LT.0.50.OR.SG100.GT.2.26) Then
C        Write (IOERR,*) ': SG100 out of range for ',
C      *      ' correlation, extrapolation occurring'
C      Else If (API.LT.14.4.OR.API.GT.58.95) Then
C        Write (IOERR,*) ': API out of range for ',
C      *      ' correlation, extrapolation occurring'
C      Else If (R.LT.2.34.OR.R.GT.2600.0) Then
C        Write (IOERR,*) ': R out of range for ',
C      *      ' correlation, extrapolation occurring'
C      End If

```

```

C
C *****
C End of validity check.
C *****
C
C     TABS = T + 460.
C
C -----
C Calculate solution gas/oil ratio.
C -----
C
C     If (API.LE.30.) Then
C       A = 0.05958
C       B = 0.7972
C       C = 1.0014
C       D = 13.1405
C     Else
C       A = 0.03150
C       B = 0.7587
C       C = 1.0937
C       D = 11.289
C     End If
C
C     RS = A * (SG100**B**P**C) * 10. ** (D*API/TABS)
C
C     If (RS.GT.R) RS = R
C   End If
C
C *****
C END OF GAS/OIL RATIO AND BUBBLE POINT CORRELATION.
C *****
C
C   Return
C   End
C *****
C       THIS SUBROUTINE CALCULATES BUBBLE POINT PRESSURES
C Written by: R.S. Trijana Kartoatmodjo           June 1991
C       * * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *
C *****
C
C   This program calculates the bubble point pressure using the common-
C   ly measured parameters in the oil field such as API gravity of stock
C   tank oil, reservoir temperature, total producing gas/oil ratio, and
C   the output from the SUBROUTINE RSTGASC.
C
C   The English system of units is used in the calculation.
C
C                               REFERENCES
C                               -----
C
C   1. Kartoatmodjo, R.S.T., Schmidt, Z.: ``New Correlations for Crude
C      Oil Physical Properties,`` SPE 023556, June 1991
C
C   2. Kartoatmodjo, R.S.T.,: ``New Correlations for Estimating
C      Hydrocarbon Liquid Physical Properties,`` M.Thesis, University
C      of Tulsa, 1990.
C
C *****
C
C                               INPUT/OUTPUT LOGICAL FILE VARIABLES
C                               -----
C
C   IOERR = Output file for error messages when input values

```


.....

CC CC

CCCC

```

      If (T.LT.75.0.OR.T.GT.320.0) Then
        Write (IOERR,*) ': T out of range for ',
        ' correlation, extrapolation occurring'
      Else If (P.LT.0.0.OR.P.GT.6040.0) Then
        Write (IOERR,*) ': P out of range for ',
        ' correlation, extrapolation occurring'
      Else If (API.LT.14.40.OR.API.GT.58.95) Then
        Write (IOERR,*) ': API out of range for ',
        ' correlation, extrapolation occurring'
      Else If (SG100.LT.0.50.OR.SG100.GT.1.67) Then
        Write (IOERR,*) ': SG100 out of range for ',
        ' correlation, extrapolation occurring'

```

```

      Else If (R.LT.2.34.OR.R.GT.2048.0) Then
        Write (IOERR,*) ': R out of range for ',
*      ' correlation, extrapolation occurring'
      End If

```

```

*****
End of validity check.
*****

```

```

-----
Calculate oil compressibility.
-----

```

```

      A = 0.83415 + 0.5002 * ALOG10(R) + 0.3613 * LOG10(API) + 0.7606
*      * ALOG10(T) - 0.35505 * ALOG10(SG100)
      CO = 10. ** A / (P*10.**6.)
End If

```

```

*****
END OF ISOTHERMAL OIL COMPRESSIBILITY CALCULATION.
*****

```

```

Return
End

```

```

*****
      THIS SUBROUTINE CALCULATES OIL FORMATION VOLUME FACTOR
Written by: R.S. Trijana Kartoatmodjo           June 1991
      * * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *
*****

```

```

This subroutine calculates oil formation volume factor using the
measured parameters in the oil fields such as API gravity of the
stock tank oil, reservoir temperature, reservoir pressure, total
producing gas/oil ratio, and the output from SUBROUTINE RSTGASC,
RSTRSOL, RSTCOMP, AND RSTBPP.

```

```

The English system of units is used in the calculation.

```

REFERENCES

1. Kartoatmodjo, R.S.T., Schmidt, Z.: ``New Correlations for Crude Oil Physical Properties,`` SPE 023556, June 1991
2. Kartoatmodjo, R.S.T.,: ``New Correlations for Estimating Hydrocarbon Liquid Physical Properties,`` M.Thesis, University of Tulsa, 1990.

```

*****

```

INPUT/OUTPUT LOGICAL FILE VARIABLES

```

IOERR = Output file for error messages when input values
        passed to the subroutine are out of range.

```

VARIABLE DESCRIPTION

```

C      *API    = API gravity of the liquid. (deg)
C      BO      = Oil formation volume factor. (bbl/STBO)
C      BOB      = Oil formation volume factor at the bubble point
C                pressure. (bbl/STBO)
C      *CO      = Oil compressibility. (psi**-1).
C                As an input value or calculated from SUBROUTINE RSTCOMP.
C      IERR      = Error code. (0=OK, 1=input variables out of range)
C      IOERR      = Output file for error messages when input values
C                passed to the subroutine are out of range.
C      *P        = Pressure. (psia)
C      *PB        = Bubble point pressure. (psia)
C                As an input value or calculated from SUBROUTINE RSTBPP.
C      *R          = Total producing gas/oil ratio. (scf/STBO)
C      *RS          = Solution gas/oil ratio. (scf/STBO)
C                As an input value or calculated from SUBROUTINE RSTRSOL.
C      SGO          = Specific gravity of oil.
C      *SG100       = Specific gravity of gas at 100 psig and T separator.
C                As an input value or calculated from SUBROUTINE RSTGASC.
C      *T          = Temperature. (deg-F)
C      A            = Dummy variables.
C      (* Indicates input variables)
C
C      *****
C
C      Subroutine RSTOFVF(T,P,API,SG100,PB,R,CO,BO,RS,IOERR,IERR)
C
C      *****
C      Check input variables for valid range.
C      *****
C
C      IERR = 0
C      If (T.LT.0.0.OR.T.GT.400.0) Then
C        Write (IOERR,*) ': Illegal input value for T'
C        IERR = 1
C      Else If (P.LT.0.0) Then
C        Write (IOERR,*) ': Illegal input value for P'
C        IERR = 1
C      Else If (API.LE.0.0.OR.API.GT.100.0) Then
C        Write (IOERR,*) ': Illegal input value for API'
C        IERR = 1
C      Else If (SG100.LT.0.50.OR.SG100.GT.3.0) Then
C        Write (IOERR,*) ': Illegal input value for SG100'
C        IERR = 1
C      Else If (PB.LT.0.0.OR.PB.GT.200000.0) Then
C        Write (IOERR,*) ': Illegal input value for PB'
C        IERR = 1
C      Else If (R.LT.0.0) Then
C        Write (IOERR,*) ': Illegal input value for R'
C        IERR = 1
C      Else If (RS.LT.0.0) Then
C        Write (IOERR,*) ': Illegal input value for RS'
C        IERR = 1
C      Else If (CO.LT.0.0) Then
C        Write (IOERR,*) ': Illegal input value for CO'
C        IERR = 1
C      End If
C      If (IERR.NE.1) Then
C
C      -----
C      Check input variables for valid correlation range.
C      -----
C
C      If (T.LT.75.0.OR.T.GT.320.0) Then
C        Write (IOERR,*) ': T out of range for ',

```

```

*      ' correlation, extrapolation occurring'
      Else If (P.LT.0.0.OR.P.GT.6040.0) Then
        Write (IOERR,*) ': P out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (API.LT.14.40.OR.API.GT.58.95) Then
        Write (IOERR,*) ': API out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (SG100.LT.0.50.OR.SG100.GT.1.67) Then
        Write (IOERR,*) ': SG100 out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (PB.LT.100.0.OR.PB.GT.6040.0) Then
        Write (IOERR,*) ': PB out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (R.LT.2.34.OR.R.GT.2048.0) Then
        Write (IOERR,*) ': R out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (RS.LT.0.0.OR.RS.GT.2048.0) Then
        Write (IOERR,*) ': RS out of range for ',
*      ' correlation, extrapolation occurring'
      Else If (CO.LT.0.00000001.OR.CO.GT.0.00010) Then
        Write (IOERR,*) ': CO out of range for ',
*      ' correlation, extrapolation occurring'
      End If

```

```

C
C *****
C End of validity check.
C *****
C
C      SGO = 141.5 / (131.5+API)
C
C      If (P.EQ.PB) Then
C
C
C
C
C
C
C

```

```

-----
Calculate oil formation volume factor at the bubble point
pressure.
-----

```

```

      BO = 0.98496 + (1.E-04) * (R**0.755*SG100**0.25*SGO**(-1.5)+
*      0.45*T) ** 1.5
      Else If (P.LT.PB.OR.RS.LT.R) Then

```

```

C
C -----
C Calculate oil formation volume factor below the bubble point
C pressure.
C -----
C

```

```

      BO = 0.98496 + (1.E-04) * (RS**0.755*SG100**0.25*SGO**(-1.5)+
*      0.45*T) ** 1.5
      Else

```

```

C
C -----
C Calculate oil formation volume factor above the bubble point
C pressure.
C -----
C

```

```

      BOB = 0.98496 + (1.E-04) * (R**0.755*SG100**0.25*SGO**(-1.5)+
*      0.45*T) ** 1.5
      BO = BOB * EXP(CO*(PB-P))
      End If
      End If

```

```

C
C *****
C END OF OIL FORMATION VOLUME FACTOR CORRELATION.
C *****
C

```

Return
End

THIS SUBROUTINE CALCULATES DEAD OIL VISCOSITY
Written by: R.S. Trijana Kartoatmodjo June 1991
* * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *

This subroutine calculates the dead oil viscosity using commonly measurable parameters in the oil fields such as API gravity of the stock tank oil, and the reservoir temperature.

The English system of units is used in the calculation.

REFERENCES

1. Kartoatmodjo, R.S.T., Schmidt, Z.: ``New Correlations for Crude Oil Physical Properties,`` SPE 023556, June 1991
2. Kartoatmodjo, R.S.T.,: ``New Correlations for Estimating Hydrocarbon Liquid Physical Properties,`` M.Thesis, University of Tulsa, 1990.

INPUT/OUTPUT LOGICAL FILE VARIABLES

IOERR = Output file for error messages when input values passed to the subroutine are out of range.

VARIABLE DESCRIPTION

*API = API gravity of the liquid. (deg)
IERR = Error code. (0=OK, 1=input variables out of range)
IOERR = Output file for error messages when input values passed to the subroutine are out of range.
*T = Temperature. (deg-F)
VISD = Dead oil viscosity. (cp)
(* Indicates input variables)

Subroutine RSTVISD(T,API,VISD,IOERR,IERR)

Check input variables for valid range.

IERR = 0
If (T.LT.0.0.OR.T.GT.600.0) Then
 Write (IOERR,*) ': Illegal input value for T'
 IERR = 1
Else If (API.LT.0.0.OR.API.GT.100.0) Then
 Write (IOERR,*) ': Illegal input value for API'
End If
If (IERR.NE.1) Then

```

C -----
C Check input variables for valid correlation range.
C -----
C
C      If (T.LT.75.0.OR.T.GT.320.0) Then
C          Write (IOERR,*) ': T out of range ',
C      *      ' , extrapolation occurring'
C      *      Else If (API.LT.14.40.OR.API.GT.58.95) Then
C          Write (IOERR,*) ': API out of range and',
C      *      ' extrapolation occurring'
C      End If
C
C      *****
C      End of validity check.
C      *****
C
C      -----
C      Calculate dead oil viscosity
C      -----
C
C      VISO = 16.0E+8 * T ** (-2.8177) * ALOG10(API) ** (5.7526*
C      *      ALOG10(T)-26.9718)
C      End If
C
C      *****
C      END OF DEAD OIL VISCOSITY CALCULATION.
C      *****
C
C      Return
C      End
C
C      *****
C      THIS SUBROUTINE CALCULATES LIVE OIL VISCOSITY
C      Written by: R.S. Trijana Kartoatmodjo      June 1991
C      * * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *
C      *****
C
C      This subroutine calculates the live oil viscosity below, at, and
C      above bubble point pressure using the commonly measured parameters
C      in the oil fields such as API gravity of stock tank oil, reservoir
C      temperature, reservoir pressure, total producing gas/oil ratio, and
C      the output from SUBROUTINE RSTRSOL, RSTBPP, and RSTVISO.
C
C      The English system of units is used in the calculation.
C
C
C      REFERENCES
C      -----
C
C      1. Kartoatmodjo, R.S.T., Schmidt, Z.: ``New Correlations for Crude
C      Oil Physical Properties,`` SPE 023556, June 1991
C
C      2. Kartoatmodjo, R.S.T.,: ``New Correlations for Estimating
C      Hydrocarbon Liquid Physical Properties,`` M.Thesis, University
C      of Tulsa, 1990.
C
C      *****
C
C      INPUT/OUTPUT LOGICAL FILE VARIABLES
C      -----
C
C      IOERR = Output file for error messages when input values

```

passed to the subroutine are out of range.

VARIABLE DESCRIPTION

*API = API gravity of the liquid. (deg)
 IERR = Error code. (0=OK, 1=input variables out of range)
 IOERR = Output file for error messages when input values
 passed to the subroutine are out of range.
 *P = Pressure. (psia)
 *PB = Bubble point pressure. (psia)
 As an input or calculated from SUBROUTINE RSTBPP.
 *RS = Solution gas/oil ratio. (scf/STBO)
 As an input or calculated from SUBROUTINE RSTRSOL.
 *R = Total producing gas/oil ratio at (scf/STBO)
 *T = Temperature. (deg-F)
 *VISD = Dead oil viscosity. (cp)
 As an input or calculated from SUBROUTINE RSTVISD.
 VISO = Live oil viscosity. (cp)
 VISB = Live oil viscosity at bubble point pressure. (cp)
 A = Dummy variable.
 (* Indicates input variables)

Subroutine RSTVISO(T,API,RS,R,P,PB,VISD,VISO,IOERR,IERR)

 Check input variables for valid range.

IERR = 0
 If (T.LT.0.0.OR.T.GT.400.0) Then
 Write (IOERR,*) ': Illegal input value for T'
 IERR = 1
 Else If (API.LT.0.0.OR.API.GT.400.0) Then
 Write (IOERR,*) ': Illegal input value for API'
 IERR = 1
 Else If (RS.LT.0.0) Then
 Write (IOERR,*) ': Illegal input value for RS'
 IERR = 1
 Else If (R.LT.0.0) Then
 Write (IOERR,*) ': Illegal input value for RS'
 IERR = 1
 Else If (P.LT.0.0) Then
 Write (IOERR,*) ': Illegal input value for P'
 IERR = 1
 Else If (PB.LT.0.0.OR.PB.GT.200000.0) Then
 Write (IOERR,*) ': Illegal input value for PB'
 IERR = 1
 Else If (VISD.LT.0.0) Then
 Write (IOERR,*) ': Illegal input value for VISD'
 IERR = 1
 End If
 If (IERR.NE.1) Then

 Check input variables for valid correlation range.

If (T.LT.75.0.OR.T.GT.320.0) Then
 Write (IOERR,*) ': T out of range ',
 * ' , extrapolation occuring'


```

      IERR = 1
    Else If (TSEP.LT.0.0.OR.TSEP.GT.400.0) Then
      Write (IOERR,*) ': Illegal input value for TSEP'
      IERR = 1
    End If
    If (IERR.NE.1) Then
C
C
C-----
C Check input variables for valid correlation range.
C-----
C
      If (T.LT.0.0.OR.T.GT.400.0) Then
        Write (IOERR,*) ': T out of range ',
*        ' extrapolation occurring'
      Else If (SGPG.LT.0.379.OR.SGPG.GE.1.709) Then
        Write (IOERR,*) ': SGPG out of range ',
*        ' extrapolation occurring'
      Else If (TSEP.LT.65.0.OR.TSEP.GT.186.0) Then
        Write (IOERR,*) ': TSEP out of range ',
*        ' extrapolation occurring'
      Else If (API.LE.17.30.OR.API.GT.54.10) Then
        Write (IOERR,*) ': API out of range ',
*        ' extrapolation occurring'
      End If
C
C *****
C
      TABS = T + 460.
      SGO = 141.5 / (131.5+API)
C
C-----
C Calculate conversion factor.
C-----
C
      CONV = 0.7264 * SGO ** 0.3202 - 0.3126 * SGPG ** (-0.02087) +
*      0.6459 * ((TSEP+460.)/TABS) ** 0.5596
      End If
C
C *****
C END OF CONVERSION FACTOR CALCULATION.
C *****
C
      Return
      End
C
C *****

```

INPUT DATA

IN SITU PRESSURE, PSIA	=	3014.7000
IN SITU TEMPERATURE, DEG. F	=	180.0000
SEPARATOR GAS SPECIFIC GRAVITY	=	0.8000
TOTAL PRODUCED GOR SCF/STB	=	500.0000
API TANK OIL GRAVITY	=	30.0000
SEPARATOR TEMPERATURE, DEG. F	=	120.0000
SEPARATOR PRESSURE, PSIA	=	124.7000

OUTPUT DATA

GAS SPECIFIC GRAVITY AT 100 PSIG	=	0.8057
FLASH SOLUTION GOR, SCF/STB	=	500.0000
BUBBLE POINT PRESSURE, PSIA	=	2387.7041
FLASH OIL FVF, bbl/STB	=	1.2750
OIL COMPRESSIBILITY, 1/PSIA	=	0.9712E-05
DEAD OIL VISCOSITY, CP	=	3.0056
LIVE OIL VISCOSITY, CP	=	0.9767
DIFFERENTIAL OIL FVF, bbl/BBL	=	1.2833
ADJUSTMENT FACTOR STB/BBL	=	0.9935
FLASH OIL FVF AT BBP, bbl/STB	=	1.2828
DIFFERENTIAL OIL FVF AT BPP, bbl/BBL	=	1.2912

INSITU PRESSURE, PSIA = 3014.700

```

      Else If (API.LT.14.40.OR.API.GT.58.95) Then
        Write (IOERR,*) ': API out of range and',
*      ' extrapolation occurring'
      Else If (RS.LT.2.00.OR.RS.GT.2048.0) Then
        Write (IOERR,*) ': RS out of range and',
*      ' extrapolation occurring'
      Else If (R.LT.2.34.OR.R.GT.2048.0) Then
        Write (IOERR,*) ': R out of range ',
*      ' extrapolation occurring'
      Else If (P.LT.0.00.OR.P.GT.7156.0) Then
        Write (IOERR,*) ': P out of range ',
*      ' extrapolation occurring'
      Else If (PB.LT.2.34.OR.PB.GT.7156.0) Then
        Write (IOERR,*) ': PB out of range ',
*      ' extrapolation occurring'
      Else If (VISD.LT.0.5.OR.VISD.GT.682.0) Then
        Write (IOERR,*) ': VISD out of range ',
*      ' extrapolation occurring'
      End If

```

```

C
C
C *****
C End of validity check.
C *****
C

```

```

C -----
C Calculate live oil viscosity below and at bubble point pressure.
C -----
C

```

```

      If (P.LT.PB.OR.RS.LT.R) Then
        A = (0.2001+0.8428*10**(-0.000845*RS)) * VISD ** (0.43+0.5165*
*      10**(-0.00081*RS))
        VISO = -0.06821 + 0.9824 * A + 0.0004034 * A ** 2
      Else If (P.EQ.PB.AND.RS.EQ.R) Then
        A = (0.2001+0.8428*10**(-0.000845*RS)) * VISD ** (0.43+0.5165*
*      10**(-0.00081*RS))
        VISO = -0.06821 + 0.9824 * A + 0.0004034 * A ** 2

```

```

C -----
C Calculate live oil viscosity above bubble point pressure.
C -----
C

```

```

      Else
        A = (0.2001+0.8428*10**(-0.000845*R)) * VISD ** (0.43+0.5165*
*      10**(-0.00081*R))
        VISB = -0.06821 + 0.9824 * A + 0.0004034 * A ** 2
        VISO = 1.00081 * VISB + 0.001127 * (P-PB) * (-0.006517*VISB**
*      1.8148+0.038*VISB**1.590)
      End If
    End If

```

```

C
C *****
C END OF LIVE OIL VISCOSITY CALCULATION.
C *****
C

```

```

      Return
    End

```

```

C *****
C THIS SUBROUTINE CALCULATES CONVERSION FACTOR FROM DIFFERENTIAL
C FLASH LIBERATION DATA TO DIFFERENTIAL LIBERATION DATA.
C Written by: R.S Trijana Kartoatmodjo June 1991
C * * TULSA UNIVERSITY ARTIFICIAL LIFT PROJECTS * *
C *****
C

```

This subroutine calculates the adjustment factor from differential liberation to flash liberation data for oil formation volume factor and gas oil ratio. The data used to calculate that conversion factor is the commonly measured parameters in the oil fields such as API gravity of stock tank oil, gravity of separator gas, separator temperature, and reservoir temperature.

The English system of units is used in the calculation.

REFERENCES

1. Kartoatmodjo, R.S.T., Schmidt, Z.: 'New Correlations for Crude Oil Physical Properties,' SPE 023556, June 1991
2. Kartoatmodjo, R.S.T.: 'New Correlations for Estimating Hydrocarbon Liquid Physical Properties,' M.Thesis, University of Tulsa, 1990.

INPUT/OUTPUT LOGICAL FILE VARIABLES

IOERR = Output file for error messages when input values passed to the subroutine are out of range.

VARIABLE DESCRIPTION

CONV = Conversion factor for BO and RS from differential liberation to flash liberation data.
 *API = API gravity of the stock tank oil. (deg)
 IERR = Error code. (0=OK, 1=input variables out of range)
 IOERR = Output file for error messages when input values passed to the subroutine are out of range.
 SGO = Specific gravity of oil.
 *SGPG = Specific gravity of produced gas.
 *T = Temperature. (deg-F)
 TABS = Absolute temperature. (deg-R)
 *TSEP = Separator temperature at which specific gravity of produced gas is reported. (deg-F)
 (* Indicates input variables)

Subroutine RSTCONV(CONV,T,SGPG,TSEP,API,IOERR,IERR)

 Check input variables for valid range.

IERR = 0
 If (API.LE.0.0.OR.API.GT.100.0) Then
 Write (IOERR,*) ': Illegal input value for API'
 IERR = 1
 Else If (SGPG.LT.0.30.OR.SGPG.GE.3.0) Then
 Write (IOERR,*) ': Illegal input value for SGPG'
 IERR = 1
 Else If (T.LT.0.0.OR.T.GT.400.0) Then
 Write (IOERR,*) ': Illegal input value for T'

INSITU TEMPERATURE, DEG. F	=	180.0000
SPECIFIC GRAVITY OF SEPARATOR GAS	=	0.8000000
TOTAL PRODUCED GOR SCF/STBO	=	500.0000
API STOCK TANK OIL GRAVITY	=	30.00000
SEPARATOR TEMPERATURE, DEG. F	=	120.0000
SEPARATOR PRESSURE, PSIA	=	124.7000

PRESS RETURN TO CONTINUE

[illegible]

1
PRESS RETURN TO CONTINUE

[illegible]

INSITU PRESSURE, PSIA	=	3014.7000
INSITU TEMPERATURE, DEG. F	=	180.0000
SPECIFIC GRAVITY OF SEPARATOR GAS	=	0.8000000
TOTAL PRODUCED GOR SCF/STBO	=	500.0000
API STOCK TANK OIL GRAVITY	=	30.00000
SEPARATOR TEMPERATURE, DEG. F	=	120.0000
SEPARATOR PRESSURE, PSIA	=	124.7000

PRESS RETURN TO CONTINUE

[illegible]

SPECIFIC GRAVITY OF GAS AT 100 PSIG	=	0.8056943
FLASH SOLUTION GOR, SCF/STB	=	500.0000
BUBBLE POINT PRESSURE, PSIA	=	2387.704
FLASH OIL FVF, bbl/STB	=	1.275005
OIL COMPRESSIBILITY, 1/PSIA	=	9.7115617E-06
DEAD OIL VISCOSITY, CP	=	3.005602
LIVE OIL VISCOSITY, CP	=	0.9766680
DIFFERENTIAL LIBERATION OIL FVF, bbl/BBL	=	1.283329

PRESS RETURN TO CONTINUE

```
ADJUSTMENT FACTOR, STB/BBL          =    0.9935141
FLASH OIL FVF AT BPP, bbl/STB      =    1.282793
DIFFERENTIAL OIL FVF AT BPP, bbl/BBL =    1.291167
PRESS RETURN TO CONTINUE
```

WHERE:

bb1 = BARREL OF OIL AT INSITU CONDITIONS

BBL = BARREL OF RESIDUAL OIL

BPP = BUBBLE POINT PRESSURE

FVF = FORMATION VOLUME FACTOR

GOR = GAS OIL RATIO

STB = BARREL OF STOCK TANK OIL

PRESS RETURN TO CONTINUE

[illegible]